

CSE7000 - INTERNSHIP

A Presentation on

TITLE OF THE INTERNSHIP

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**Presidency School of Computer Science
and Engineering**

Name of the Program: B. Tech (CSE)

Name of the HoD: Dr. Nagaraja S R

Name of the Program Internship Coordinator: Ms. Vidhya Rengasamy

Name of the School Internship Coordinators: Dr. Bhuvaneshwari Patil



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About Company or Organization

AI GPU Summer Internship Program

- Conducted by **Presidency University – School of AI and Advanced Computing (GPU Computing Lab)**.
- Focused on Artificial Intelligence, High-Performance Computing (HPC), GPU programming, and modern AI application development.
- Provided hands-on training in deep learning, Large Language Models (LLMs), computer vision, NLP, and GPU-accelerated computing.
- Students gained practical experience using NVIDIA H200 Tensor Core GPUs and industry-standard AI development tools.
- The internship emphasized solving real-world problems through end-to-end AI project development.



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Working domain or the technology

Working Domain

- PulseMind AI is developed in the **Healthcare Technology (HealthTech)** domain. It combines Artificial Intelligence, Cloud Computing, Medical Informatics, and Full Stack Development to improve healthcare services.
- The platform is designed to assist **patients, doctors, hospitals, radiologists**, by providing intelligent healthcare solutions such as medical report analysis, disease prediction, emergency triage, and personalized health recommendations.



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Technologies Used:

Frontend : React.js ,TypeScript , Vite ,Tailwind CSS , HTML5 ,CSS3 ,JavaScript

Backend : FastAPI ,Python ,REST APIs

Artificial Intelligence :Large Language Models (LLMs) , Medical NLP ,Explainable AI (XAI) ,Clinical Decision Support ,AI Risk Prediction ,Disease Progression Simulation

Database: Firebase Firestore , JSON Local Storage (Fallback)

Medical Processing : OCR (Optical Character Recognition) , Medical Image Classification ,X-Ray Analysis ,Clinical Report Analysis

Data Visualization: Interactive Charts ,Health Dashboards ,Risk Graphs ,Disease Analytics ,Progress Monitoring

Development Tools: Git ,GitHub ,VS Code ,Postman

Objectives of the work

- To develop an AI-powered intelligent healthcare platform.
- To assist healthcare professionals in clinical decision-making.
- To predict disease risks using Artificial Intelligence.
- To analyze medical reports and medical images automatically.
- To provide AI-based emergency triage and first-aid guidance.
- To simulate disease progression under different treatment scenarios.
- To manage medications, appointments, nutrition, and patient health records.
- To identify hereditary disease risks through family health analysis.
- To improve healthcare accessibility using modern web technologies and AI.
- To build an integrated, scalable, and user-friendly digital healthcare solution suitable for real-world clinical applications.



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Methodology Used / Course Description

1. Requirement Analysis

- Identified major healthcare challenges such as delayed diagnosis, manual report analysis, emergency response, and patient management. Collected functional and non-functional requirements. Designed the overall workflow and healthcare modules.

2. System Design

- Designed the user interface (UI/UX) for an intuitive healthcare dashboard. Planned the frontend, backend, AI modules, and database architecture. Created modular components for each healthcare feature.

3. Frontend Development

- React.js ,TypeScript ,Tailwind CSS ,Vite ,
- **Features developed include:** Dashboard ,AI Clinical Assistant ,Disease Progression Simulator ,Medical Imaging Emergency Triage ,Medication Management ,Nutrition Planner ,Appointment Scheduler ,Research Agent ,Family Health Analysis

4. Backend Development:

Developed RESTful APIs using: FastAPI (Python) ,Pydantic ,REST API architecture

The backend handles: AI request processing ,Medical report analysis ,Patient data management ,Prediction models ,File uploads ,Authentication ,API communication.

5. Artificial Intelligence Integration:

Integrated multiple AI technologies: Large Language Models (LLMs) ,Natural Language Processing (NLP) ,OCR for medical report extraction ,Disease prediction algorithms ,Explainable AI (XAI) ,Clinical decision support.

6. Database Management

Used Firebase Firestore (with local JSON fallback) for storing: Patient records ,Medical reports ,Medication schedules ,Appointment information ,AI analysis history ,User profiles.

7. Testing and Validation

Performed: Functional testing ,UI testing ,API testing ,AI output validation ,Performance testing ,Error handling
Cross-browser compatibility testing.

Course Description

- **PulseMind AI** : is an AI-powered Intelligent Clinical Healthcare and Diagnostic Intelligence Platform developed to enhance healthcare services through modern digital technologies. The project integrates **Artificial Intelligence (AI), Machine Learning (ML), Explainable AI (XAI), OCR, and Full-Stack Web Development** to provide intelligent healthcare solutions.
- The platform includes modules for **disease risk prediction, AI-powered medical assistance, medical report and image analysis, emergency triage, nutrition planning, medication reminders, appointment management, family health analysis, disease progression simulation, and healthcare analytics**. It enables patients and healthcare professionals to make informed clinical decisions through personalized recommendations and real-time health insights.
- The application is developed using **React.js, TypeScript, Tailwind CSS, FastAPI, Python, and Firebase Firestore**, providing a secure, scalable, and user-friendly healthcare system that supports preventive care, efficient patient management, and intelligent clinical decision-making.



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Conclusion

- PulseMind AI successfully demonstrates the application of **Artificial Intelligence, Machine Learning, and Full-Stack Web Development** in creating an intelligent healthcare platform. The system integrates multiple healthcare services, including disease prediction, medical report analysis, medical imaging, emergency triage, nutrition planning, medication management, appointment scheduling, and AI-assisted clinical support, into a single user-friendly application.
- The project provides accurate health insights, supports informed clinical decision-making, and improves patient care through intelligent automation and real-time analytics. Developed using **React.js, TypeScript, FastAPI, Python, and Firebase Firestore**, the platform is secure, scalable, and easy to use. Overall, PulseMind AI demonstrates how modern AI technologies can be effectively applied to address real-world healthcare challenges and provides a strong foundation for future enhancements such as IoT integration, wearable device support, telemedicine, and Electronic Health Record (EHR) systems.



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Q&A



Thank you !!



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